Lab 8 Pair Programming – More Objects: Inheritance and Polymorphism

Aim

This week is focused on the object inheritance, overriding methods, and polymorphism. You will start using Pair Programming to learn more from each other.

Pair Programming Lab:

1. The Lab TA will create a pair within the group.
2. Talk to your pair, who is going to be the Driver first.
3. Complete Task 1 with Pair Programming (30-45 minutes).
4. Show the TA the results of your works in Task 1.
5. Switch roles.
6. Complete Task 2 with Pair Programming (30-45 minutes).
7. Report to TA after you have completed Task 2.

Pair Programming Lab 8 Task 1: Books in Home Library

- Develop a Java program that represents books, in a home library.
- Create a **Book** class, that stores the author, title, number of books in the home library, and other instance/class variables that you can think of.
- Add instance methods and static methods (functions) to access or modify the Book instances.
- Create two subclasses: **Fiction** and **NonFiction**, both inheriting from the Book class.
- Add instance variables and methods specific to those subclasses.
- Implement overriding on methods such as details(), to display the details of each specific book.
- In a client class, create an array of Book objects containing instances of both Fiction and NonFiction books.
- Demonstrate polymorphism by iterating through the array, displaying the details (title, author, and any specific attributes for each book), or calling your other overidden methods.

Pair Programming Lab 8 Task 2: Restaurant Management System

- Develop a Java program that manages different types of food items in a restaurant.
- Create a **Food** class, that stores the food name, price, total number of food items in the restaurant, and other instance/class variables that you can think of.
- Add instance methods and static methods (functions) to access or modify the Food instances.
- Create two subclasses: **MainCourse** and **Dessert**, both inheriting from the Food class.
- Add instance variables and methods specific to those subclasses.
- Implement overriding on methods such as serve(), to demonstate how each food item is served.
- In a client class, create an array of Food objects containing instances of both MainCourse and Dessert food.
- Demonstrate polymorphism by iterating through the array, displaying how a particular food is served, or calling your other overidden methods.

This is the end of CPT111-2324 Pair Programming Lab 8 Task Sheet.